

Online Supplement to “Optimal Model Averaging for Kernel Ridge Regression and Its Divide-and-Conquer Strategy”

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Part A: Justification and Interpretation of Some Conditions

In this section, we provide some explanations and discussions on Conditions 2, 6 and 10.

A.1. Discussion on Condition 2

Recall that $\xi_N = \inf_{\mathbf{w} \in \mathcal{W}} R(\mathbf{w}) = \inf_{\mathbf{w} \in \mathcal{W}} E(\|f_0(\mathbf{X}) - \hat{f}_{\mathbf{w}}(\mathbf{X})\|_N^2)$ is the minimum risk among all model averaging estimators and $R_m = E(\|f_0(\mathbf{X}) - \hat{f}_m(\mathbf{X})\|^2)$. Condition 2 requires the term $M^{2/\gamma} \bar{R}^{1/2} N^{-1}$ to be asymptotically negligible relative to the oracle risk ξ_N . We next discuss Condition 2 under several representative cases.

(1) First, suppose that all candidate models are correctly specified, that is, $f_0 \in \mathcal{H}_m$ for all $m \in \{1, \dots, M\}$. This may occur when the candidate models are constructed from the same RKHS induced by a common kernel, but with different tuning parameters $\lambda_1, \dots, \lambda_M$. In this case, $\mathcal{H}_1 = \dots = \mathcal{H}_M = \mathcal{H}$ and $f_0 \in \mathcal{H}$, so all candidate KRR models are correctly specified, although their finite-sample performance may differ due to different choices of the tuning parameter. In general, the natural choices of the tuning parameters λ_m are around the rate-optimal order which depends on the specific kernel. To illustrate the reasonableness of Condition 2, we consider two representative types of kernels.

(i) For the kernels with polynomially decaying eigenvalues and decay parameter α ($\alpha > 1/2$), the optimal order of tuning parameters is $\Theta(N^{-2\alpha/(2\alpha+1)})$ (Zhang et al. 2015). In this situation, we can assume that the order of R_m/N is $O(N^{-2\alpha/(2\alpha+1)})$ uniformly for $m \in \{1, \dots, M\}$ according to the classical KRR theory (see, e.g., Theorem 1 in Zhang et al. (2025) and its corollary). Given that all candidate models have the same risk order, it is natural to assume that the oracle risk ξ_N has the same order $N^{-2\alpha/(2\alpha+1)}$. Similarly, we have $\bar{R}/N = \max_{1 \leq m \leq M} R_m/N = O(N^{-2\alpha/(2\alpha+1)})$. Further, we can derive that $M^{2/\gamma} \bar{R}^{1/2} N^{-1}/\xi_N = O(M^{2/\gamma} N^{-1/(4\alpha+2)})$. Hence, Condition 2 is satisfied if $M = o(N^{\gamma/(8\alpha+4)})$.

(ii) For the kernels with exponentially decaying eigenvalues whose j th eigenvalue satisfies a sub-Gaussian decay condition of the form $\exp(-cj^2)$, such as the Gaussian kernel, Zhang et al. (2015) suggested that the optimal order of the tuning parameters is $\Theta(N^{-1})$. Following the related theory (see, e.g., Corollary 5 in Zhang et al. (2015)), we can assume that $R_m/N = O(\sqrt{\log N}/N)$ uniformly for $m = 1, \dots, M$, which implies $\bar{R}/N = O(\sqrt{\log N}/N)$. Further, the oracle risk ξ_N is expected to follow the same rate $\sqrt{\log N}/N$. Substituting these orders into Condition 2, we obtain $M^{2/\gamma} \bar{R}^{1/2} N^{-1}/\xi_N = O(M^{2/\gamma} (\log N)^{-1/4})$. Hence, in the exponential eigenvalue decay case, Condition 2 holds provided that $M = o\{(\log N)^{\gamma/8}\}$. The above discussion shows that, when all candidate models are correctly specified with the optimal choices of the tuning parameters, Condition 2 can be satisfied under suitable growth rates of the number of candidate models M .

(2) Second, suppose that all candidate models are misspecified, that is, $f_0 \notin \mathcal{H}_m$ for all $m = 1, \dots, M$. This is a common situation in practice, since the true data-generating mechanism is usually unknown and may not be contained in any prespecified candidate Hilbert space. In this case, the limiting target of $\hat{f}_m$ is generally not f_0 , but a pseudo-true function f_m^* (see Section 4.2 for details). For any $\mathbf{w} \in \mathcal{W}$, define $f_{\mathbf{w}}^*(\mathbf{x}) = \sum_{m=1}^M w_m f_m^*(\mathbf{x})$. Then the risk $R(\mathbf{w})$ can be decomposed into the approximation error and the estimation error. Specifically,

$$f_0(\mathbf{X}) - \hat{f}_{\mathbf{w}}(\mathbf{X}) = f_0(\mathbf{X}) - f_{\mathbf{w}}^*(\mathbf{X}) + f_{\mathbf{w}}^*(\mathbf{X}) - \hat{f}_{\mathbf{w}}(\mathbf{X}).$$

Hence, $R(\mathbf{w}) = A_N(\mathbf{w}) + B_N(\mathbf{w}) + 2D_N(\mathbf{w})$, where $A_N(\mathbf{w}) = \|f_0(\mathbf{X}) - f_{\mathbf{w}}^*(\mathbf{X})\|_N^2$ is the approximation error caused by model misspecification, $B_N(\mathbf{w}) = E(\|\hat{f}_{\mathbf{w}}(\mathbf{X}) - f_{\mathbf{w}}^*(\mathbf{X})\|_N^2)$ is the estimation error, and $D_N(\mathbf{w}) = E(N^{-1}\{f_0(\mathbf{X}) - f_{\mathbf{w}}^*(\mathbf{X})\}^T \{f_{\mathbf{w}}^*(\mathbf{X}) - \hat{f}_{\mathbf{w}}(\mathbf{X})\})$ is the cross term. Let $A_N^* = \inf_{\mathbf{w} \in \mathcal{W}} A_N(\mathbf{w}) = \inf_{\mathbf{w} \in \mathcal{W}} \|f_0(\mathbf{X}) - f_{\mathbf{w}}^*(\mathbf{X})\|_N^2$ be the minimum approximation error induced by the pseudo-true functions.

We focus on the case where the approximation error caused by model misspecification cannot be eliminated by model averaging. That is, there exists a constant $c > 0$ such that

$$A_N^* = \inf_{\mathbf{w} \in \mathcal{W}} \|f_0(\mathbf{X}) - f_{\mathbf{w}}^*(\mathbf{X})\|_N^2 \geq c.$$

Such a case is practically relevant, since the prespecified candidate Hilbert spaces may still be unable to fully capture the true regression function. One typical example is that all candidate models are constructed without some important covariates, leading to a non-negligible approximation error. For the estimation error, according to the KRR theory (see, e.g., Theorem 2 in Zhang et al. (2015)), the candidate estimators can be assumed to satisfy $\|\hat{f}_m(\mathbf{X}) - f_m^*(\mathbf{X})\|_N^2 = O_p(a_N^2)$ uniformly for $m \in \{1, \dots, M\}$ with $a_N \rightarrow 0$ as $N \rightarrow \infty$. The sequence a_N may vary across different scenarios; see Remark 3 for detailed discussion. Then, for weights in the simplex, the estimation error $B_N(\mathbf{w})$ is asymptotically negligible uniformly over $\mathbf{w} \in \mathcal{W}$. Thus ξ_N is mainly determined by A_N^* . Here and hereafter, $u_N \asymp v_N$ means that u_N/v_N is bounded away from zero and infinity. Given that $A_N^* \geq c$, the oracle risk satisfies $\xi_N \asymp 1$.

Moreover, in this misspecified setting, we can assume $\bar{R} = O(N)$. Substituting $\xi_N \asymp 1$ and $\bar{R} = O(N)$ into Condition 2, we obtain

$$\frac{M^{2/\gamma} \bar{R}^{1/2} N^{-1}}{\xi_N} = O\left(M^{2/\gamma} N^{-1/2}\right).$$

Therefore, Condition 2 is satisfied if $M = o(N^{\gamma/4})$. This shows that Condition 2 is reasonable and can be satisfied under model misspecification.

(3) Finally, consider the mixed case where both correctly specified and misspecified candidate models are included. That is, there exists at least one m_0 such that $f_0 \in \mathcal{H}_{m_0}$, while there also exists some m_1 such that $f_0 \notin \mathcal{H}_{m_1}$. In this case, the order of the oracle risk ξ_N is mainly determined by the correctly specified candidate models. For example, if the correctly specified candidate model attains the normalized risk rate r_N , then we have $\xi_N \leq R(e_{m_0}) = O(r_N)$, where e_{m_0} is a M -dimension vector with the m_0 th element being one and the remaining elements being zero. Thus, it is natural to regard ξ_N as being of the same order as r_N in this case.

On the other hand, the quantity $\bar{R} = \max_{1 \leq m \leq M} R_m$ is affected by the misspecified candidate models. For a misspecified model with a non-negligible approximation error, its normalized risk R_m/N is typically of constant order. Therefore, $\bar{R} = \max_{1 \leq m \leq M} R_m = O(N)$. Substituting $\bar{R} = O(N)$ and $\xi_N \asymp r_N$ into Condition 2, we obtain

$$\frac{M^{2/\gamma} \bar{R}^{1/2} N^{-1}}{\xi_N} = O\left(\frac{M^{2/\gamma} N^{-1/2}}{r_N}\right).$$

When r_N decreases slower than $N^{-1/2}$, this condition holds as long as M takes an appropriate rate. However, for many Hilbert spaces induced by commonly used kernels, r_N typically decreases faster than $N^{-1/2}$. For example, $r_N = N^{-2\alpha/(2\alpha+1)}$ for polynomially decaying kernels with decay parameter $\alpha > 1/2$, and $r_N = \sqrt{\log N}/N$ for exponentially decaying kernels. In these cases, Condition 2 may fail even when M is fixed.

Therefore, Condition 2 may exclude this mixed case. Nevertheless, this does not imply that the proposed method performs poorly in such cases. Theorem 3 establishes the consistency of the estimated weights without requiring Condition 2, that is, the data-driven weights can still converge to the optimal weight. This also provides the theoretical guarantee for the performance of the proposed method.

A.2. Discussion on Condition 6

Condition 6 is similar in spirit to Condition 2. It requires the remainder terms $M^{2/\gamma} \bar{R}^{1/2} N^{-1}$ and $\zeta_N = \max\{a_N, MN^{-1/2}\}$ to be asymptotically negligible relative to the oracle risk $\xi_N = \inf_{\mathbf{w} \in \mathcal{W}} R(\mathbf{w})$. As discussed for Condition 2, this requirement can be satisfied in Case (2), where $\xi_N \asymp 1$, provided that the number of candidate models M grows at an appropriate rate. The detailed verification is similar to the discussion of Condition 2 and is therefore omitted.

For Cases (1)-(i) and (1)-(ii), the situation is slightly different. In these cases, we have $a_N = N^{-\alpha/(2\alpha+1)}$ for Case (1)-(i) and $a_N = N^{-1/2}(\log N)^{1/4}$ for Case (1)-(ii). Then the requirement $\zeta_N/\xi_N = o(1)$ may no longer be satisfied. Therefore, Condition 6 may fail in Cases (1)-(i) and (1)-(ii). Similarly, as discussed for Condition 2, Condition 6 may also fail in Case (3). However, this does not indicate poor performance of the proposed CVMA method in these cases. It only indicates that Condition 6 is a sufficient condition for establishing the relative asymptotic optimality result, rather than a necessary condition for the proposed method to perform well. Theorem 5 provides a complementary guarantee, which does not rely on Condition 6 and ensures that the estimated weights converge to the optimal weights. Therefore, even when Condition 6 fails in Cases (1)-(i), (1)-(ii), and (3), the proposed method remains theoretically justified from the perspective of weight consistency.

A.3. Discussion on Condition 10

Condition 10 plays a similar role to Conditions 2 and 6. Compared with Condition 6, Condition 10 additionally involves the number of machines L , which reflects the extra approximation error introduced by the divide-and-conquer implementation.

As discussed for Conditions 2 and 6, this type of requirement can be satisfied in Case (2), where $\xi_N \asymp 1$. In this case, Condition 10 essentially requires

$$ML^{1/2}\bar{R}_{\max}^{1/2}N^{-1} \rightarrow 0 \quad \text{and} \quad \zeta_n = \max\{a_n, M^{1/2}N^{-1/2}\} \rightarrow 0,$$

which can hold provided that the numbers of candidate models and machines grow at suitable rates. In particular, L should not grow too fast so that each local sample size $n = N/L$ still tends to infinity and the local approximation error a_n vanishes. The detailed verification is similar to the discussion of Conditions 2 and 6 and is therefore omitted.

For Cases (1)-(i), (1)-(ii) and (3), Condition 10 may fail because the oracle risk ξ_N can be relatively small, so the above remainder terms may no longer be negligible relative to ξ_N . However, similar to the CVMA method, this possible failure should not be interpreted as poor performance of the proposed DCMA method. It only should be viewed as a technical sufficient condition for the relative-risk analysis rather than a necessary condition for the effectiveness of the proposed distributed model averaging procedure. This viewpoint is also supported by the numerical results in Section 6, in which the proposed DCMA method shows satisfactory performance across different simulation settings.

Part B: Additional Numerical Results

Example 2. This example considers the true regression function:

$$f_0(\mathbf{x}_i) = 2x_{i1} - x_{i2} + 1.5(x_{i3}^2 - 1/3) + 2x_{i1}x_{i2} + x_{i4}x_{i5}, \quad i = 1, \dots, N. \quad (\text{S.1})$$

In this example, the true regression function is a polynomial of degree at most two. Therefore, the candidate set contains correctly specified candidate spaces. The other settings are the same as in Example 1.

The AMSPE values of different procedures for Example 2 are displayed in Figure S1. Since the candidate set contains a correctly specified quadratic polynomial model, GCV and CV become highly competitive, particularly when $N = 400$. Also, MTMA-T also yields the smallest or nearly the smallest AMSPE in most settings. Additionally, CVMA-5 and CVMA-10 still exhibit very similar performance, and they remain close to the best-performing method across all settings. MTMA-E performs slightly worse than MTMA-T due to the uncertainty in estimating the error variance, but remains competitive as the sample size and SNR increase. Finally, EW performs well under relatively low SNRs when $N = 200$, but it gradually loses its advantage as the SNR increases. Overall, the results from Examples 1 and 2 confirm the stable predictive performance of MTMA and CVMA.

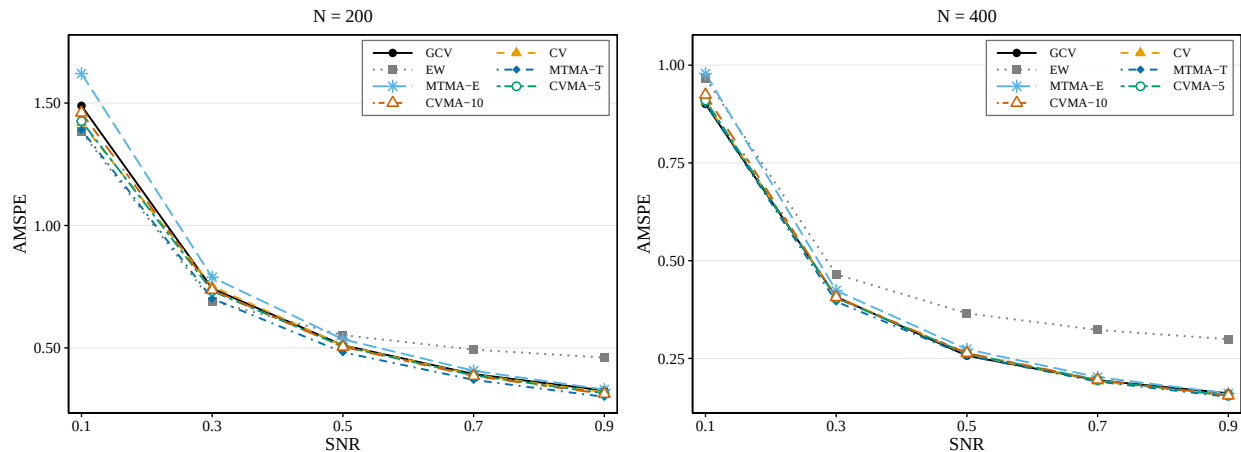


Figure S1 The simulation results in terms of AMSPE for Example 2.

Example 4. We adopt the same quadratic polynomial regression function (S.1) as in Example 2. This example examines the performance of the distributed procedures when a correct candidate model is available with large sample sizes.

We present the simulation results for Example 4 in Figure S2. It can be seen that Q-DCMA-T and Q-DCMA-E yield the smallest or nearly the smallest AMSPE in almost all cases. Further, Q-DCMA-T and Q-DCMA-E are nearly indistinguishable, indicating that estimating the error variance is reliable in the settings of Example 4. DGCV also remains close to the Q-DCMA procedures because it can benefit from the correctly specified candidate model. Meanwhile, DCMA maintains competitive predictive performance across the different sample sizes and numbers of machines. This finding also illustrates that, although the presence of some correctly specified candidates may make the sufficient condition on ξ_N for the asymptotic optimality of DCMA difficult to satisfy, DCMA can still exhibit favorable finite-sample performance in such settings. In contrast, DEW performs considerably worse. The computation-time patterns are similar to those in Example 3: the distributed procedures become substantially faster as L increases and are much more computationally efficient than the available full-sample benchmarks. For instance, when $N = 2^{11}$ and $L = 8$, Q-DCMA-E requires only about 6.1 seconds, compared with 138.7 seconds for Full MTMA-E and 1101.0 seconds for Full CVMA-10, while its AMSPE remains very close to those of the two full-sample benchmarks. As a whole, the results from Examples 3 and 4 support the predictive effectiveness and computational scalability of the proposed distributed model averaging methods.

Part C: Proofs of the Theoretical Results

C.1. Proof of Theorem 1

Observe that $\hat{f}_{\mathbf{w}}(X) = \mathbf{H}(\mathbf{w})Y = \mathbf{H}(\mathbf{w})f_0(X) + \mathbf{H}(\mathbf{w})\epsilon$ with $\epsilon = (\epsilon_1, \dots, \epsilon_N)^T$. So we have

$$R(\mathbf{w}) = E(\|f_0(X) - \hat{f}_{\mathbf{w}}(X)\|_N^2) = N^{-1}[\|(\mathbf{H}(\mathbf{w}) - \mathbf{I}_N)f_0(X)\|^2 + \sigma^2 \text{tr}(\mathbf{H}^T(\mathbf{w})\mathbf{H}(\mathbf{w}))].$$

Similarly, we also have

$$\begin{aligned} E(\|Y - \hat{f}_{\mathbf{w}}(X)\|^2) &= \|(\mathbf{H}(\mathbf{w}) - \mathbf{I}_N)f_0(X)\|^2 + \sigma^2 \text{tr}((\mathbf{H}(\mathbf{w}) - \mathbf{I}_N)^T(\mathbf{H}(\mathbf{w}) - \mathbf{I}_N)) \\ &= \|(\mathbf{H}(\mathbf{w}) - \mathbf{I}_N)f_0(X)\|^2 + \sigma^2(N - 2\text{tr}(\mathbf{H}(\mathbf{w})) + \text{tr}(\mathbf{H}^T(\mathbf{w})\mathbf{H}(\mathbf{w}))). \end{aligned}$$

Finally, using the above equation, we can obtain that

$$E(C_N(\mathbf{w})) = N^{-1}E(\|Y - \hat{f}_{\mathbf{w}}(X)\|^2) + 2N^{-1}\sigma^2 \text{tr}(\mathbf{H}(\mathbf{w})) = R(\mathbf{w}) + \sigma^2.$$

This completes the proof of Theorem 1.

C.2. Proof of Theorem 2

Let $C^*(\mathbf{w}) = C(\mathbf{w}) - \|\epsilon\|^2$. Thus, $\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathcal{W}} C(\mathbf{w}) = \arg \min_{\mathbf{w} \in \mathcal{W}} C_N^*(\mathbf{w})$, where $C_N^*(\mathbf{w}) = N^{-1}C^*(\mathbf{w})$. To verify Theorem 2, it suffices to prove that

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{C_N^*(\mathbf{w}) - L(\mathbf{w})}{R(\mathbf{w})} \right| \xrightarrow{P} 0 \quad (\text{S.2})$$

and

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{L(\mathbf{w}) - R(\mathbf{w})}{R(\mathbf{w})} \right| \xrightarrow{P} 0. \quad (\text{S.3})$$

First, we focus on the proof of (S.2). Note that

$$\begin{aligned} \sup_{\mathbf{w} \in \mathcal{W}} |C_N^*(\mathbf{w}) - L(\mathbf{w})| &= \sup_{\mathbf{w} \in \mathcal{W}} N^{-1}|2\epsilon^T(f_0(X) - \hat{f}_{\mathbf{w}}(X)) + 2\sigma^2 \text{tr}(\mathbf{H}(\mathbf{w}))| \\ &\leq \sup_{\mathbf{w} \in \mathcal{W}} N^{-1}|2\epsilon^T(f_0(X) - \mathbf{H}(\mathbf{w})f_0(X))| + \sup_{\mathbf{w} \in \mathcal{W}} N^{-1}|2\epsilon^T \mathbf{H}(\mathbf{w})\epsilon - 2\sigma^2 \text{tr}(\mathbf{H}(\mathbf{w}))| \triangleq T_{11} + T_{12}. \end{aligned}$$

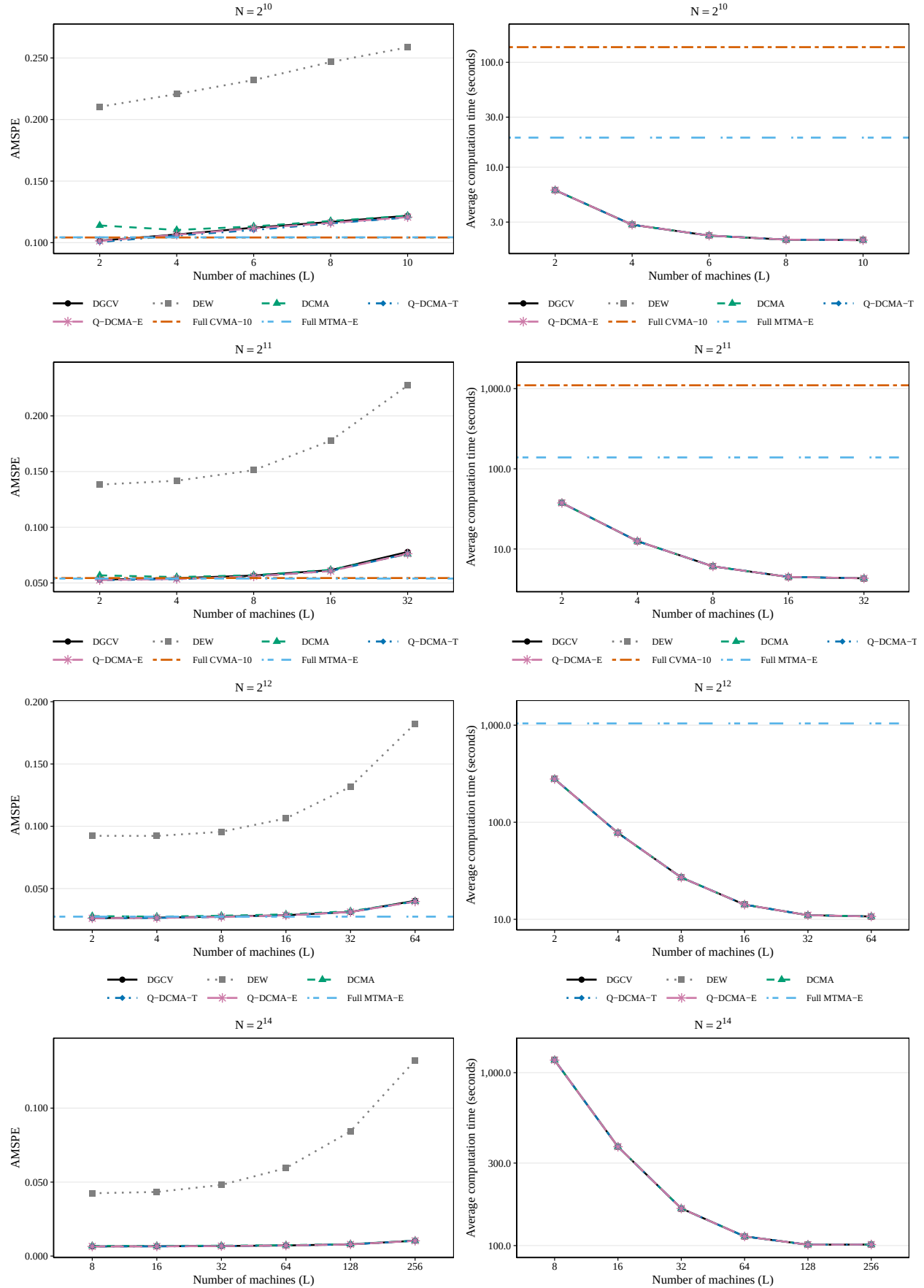


Figure S2 The simulation results in terms of AMSPE for Example 4.

For the term T_{11} , by Theorem 2 in Whittle (1960) and Condition 1, there exists some constant $\gamma \geq 2$ such that

$$\begin{aligned} E \left| \sup_{\mathbf{w} \in \mathcal{W}} |2\boldsymbol{\epsilon}^T(f_0(\mathbf{X}) - \mathbf{H}(\mathbf{w})f_0(\mathbf{X}))| \right|^\gamma &\leq E \left| \sup_{\mathbf{w} \in \mathcal{W}} \sum_{m=1}^M w_m |2\boldsymbol{\epsilon}^T(f_0(\mathbf{X}) - \mathbf{H}_m f_0(\mathbf{X}))| \right|^\gamma \\ &\leq \sum_{m=1}^M E |2\boldsymbol{\epsilon}^T(f_0(\mathbf{X}) - \mathbf{H}_m f_0(\mathbf{X}))|^\gamma \leq C \sum_{m=1}^M \|f_0(\mathbf{X}) - \mathbf{H}_m f_0(\mathbf{X})\|^\gamma \leq CM\bar{R}^{\gamma/2}. \end{aligned}$$

Thus, we have $T_{11} = O_p(N^{-1}M^{1/\gamma}\bar{R}^{1/2})$. Similarly, for the term T_{12} , by Theorem 2 in Whittle (1960) and Condition 1, we can derive that

$$\begin{aligned} E \left| \sup_{\mathbf{w} \in \mathcal{W}} |2\boldsymbol{\epsilon}^T \mathbf{H}(\mathbf{w})\boldsymbol{\epsilon} - 2\sigma^2 \text{tr}(\mathbf{H}(\mathbf{w}))| \right|^\gamma &\leq E \left| \sup_{\mathbf{w} \in \mathcal{W}} \sum_{m=1}^M w_m |2\boldsymbol{\epsilon}^T \mathbf{H}_m \boldsymbol{\epsilon} - 2\sigma^2 \text{tr}(\mathbf{H}_m)| \right|^\gamma \\ &\leq \sum_{m=1}^M E |2\boldsymbol{\epsilon}^T \mathbf{H}_m \boldsymbol{\epsilon} - 2\sigma^2 \text{tr}(\mathbf{H}_m)|^\gamma \leq C \sum_{m=1}^M (\text{tr}(\mathbf{H}_m^T \mathbf{H}_m))^\gamma \leq CM\bar{R}^{\gamma/2}. \end{aligned}$$

Then, we have $T_{12} = O_p(N^{-1}M^{1/\gamma}\bar{R}^{1/2})$. Further, using Condition 2,

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{C_N^*(\mathbf{w}) - L(\mathbf{w})}{R(\mathbf{w})} \right| \leq \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |C_N^*(\mathbf{w}) - L(\mathbf{w})| = O_p(\xi_N^{-1}N^{-1}M^{1/\gamma}\bar{R}^{1/2}) = o_p(1).$$

This proves (S.2).

Moreover, to prove (S.3), we decompose

$$\begin{aligned} &\sup_{\mathbf{w} \in \mathcal{W}} |L(\mathbf{w}) - R(\mathbf{w})| \\ &= \sup_{\mathbf{w} \in \mathcal{W}} N^{-1} \|\mathbf{f}_0(\mathbf{X}) - \mathbf{H}(\mathbf{w})\mathbf{Y}\|^2 - \|\mathbf{H}(\mathbf{w})\mathbf{f}_0(\mathbf{X}) - \mathbf{f}_0(\mathbf{X})\|^2 - \sigma^2 \text{tr}(\mathbf{H}^T(\mathbf{w})\mathbf{H}(\mathbf{w})) \\ &\leq \sup_{\mathbf{w} \in \mathcal{W}} N^{-1} |2\boldsymbol{\epsilon}^T \mathbf{H}^T(\mathbf{w})(\mathbf{H}(\mathbf{w})\mathbf{f}_0(\mathbf{X}) - \mathbf{f}_0(\mathbf{X}))| + \sup_{\mathbf{w} \in \mathcal{W}} N^{-1} |\boldsymbol{\epsilon}^T \mathbf{H}^T(\mathbf{w})\mathbf{H}(\mathbf{w})\boldsymbol{\epsilon} - \sigma^2 \text{tr}(\mathbf{H}(\mathbf{w}))| \triangleq T_{21} + T_{22}. \end{aligned}$$

Observe that $\lambda_{\max}(\mathbf{H}_m \mathbf{H}_m^T) = \lambda_{\max}((\mathbf{K}_m + \lambda_m N \mathbf{I}_N)^{-2} \mathbf{K}_m^2) \leq 1$. Then, by Theorem 2 in Whittle (1960) and Condition 1, we have

$$\begin{aligned} &E \left| \sup_{\mathbf{w} \in \mathcal{W}} |2\boldsymbol{\epsilon}^T \mathbf{H}^T(\mathbf{w})(\mathbf{H}(\mathbf{w})\mathbf{f}_0(\mathbf{X}) - \mathbf{f}_0(\mathbf{X}))| \right|^\gamma \\ &\leq E \left| \sup_{\mathbf{w} \in \mathcal{W}} 2 \sum_{m=1}^M w_m \sum_{l=1}^M w_l |\boldsymbol{\epsilon}^T \mathbf{H}_m^T (\mathbf{H}_l \mathbf{f}_0(\mathbf{X}) - \mathbf{f}_0(\mathbf{X}))| \right|^\gamma \leq C \sum_{m=1}^M \sum_{l=1}^M \|\mathbf{H}_m^T (\mathbf{H}_l \mathbf{f}_0(\mathbf{X}) - \mathbf{f}_0(\mathbf{X}))\|^\gamma \\ &\leq C \sum_{m=1}^M \sum_{l=1}^M (\lambda_{\max}(\mathbf{H}_m \mathbf{H}_m^T))^{\gamma/2} \|\mathbf{H}_l \mathbf{f}_0(\mathbf{X}) - \mathbf{f}_0(\mathbf{X})\|^\gamma \leq CM^2 \bar{R}^{\gamma/2}. \end{aligned}$$

Thus, we can deduce that $T_{21} = O_p(M^{2/\gamma}\bar{R}^{1/2})$. Note that

$$|\boldsymbol{\epsilon}^T \mathbf{H}^T(\mathbf{w})\mathbf{H}(\mathbf{w})\boldsymbol{\epsilon} - \sigma^2 \text{tr}(\mathbf{H}(\mathbf{w}))| \leq \sum_{m=1}^M \sum_{l=1}^M w_m w_l |\boldsymbol{\epsilon}^T \mathbf{H}_m^T \mathbf{H}_l \boldsymbol{\epsilon} - \sigma^2 \text{tr}(\mathbf{H}_m^T \mathbf{H}_l)|.$$

Similarly, by Theorem 2 in Whittle (1960) and Condition 1, we have

$$E \left| \sup_{\mathbf{w} \in \mathcal{W}} |\boldsymbol{\epsilon}^T \mathbf{H}^T(\mathbf{w})\mathbf{H}(\mathbf{w})\boldsymbol{\epsilon} - \sigma^2 \text{tr}(\mathbf{H}(\mathbf{w}))| \right|^\gamma$$

$$\begin{aligned} &\leq \sum_{m=1}^M \sum_{l=1}^M E |\epsilon^T \mathbf{H}_m^T \mathbf{H}_l \epsilon - \sigma^2 \text{tr}(\mathbf{H}_m^T \mathbf{H}_l)|^\gamma \leq C \sum_{m=1}^M \sum_{l=1}^M (\text{tr}(\mathbf{H}_m \mathbf{H}_m^T \mathbf{H}_l \mathbf{H}_l^T))^{\gamma/2} \\ &\leq C \sum_{m=1}^M \sum_{l=1}^M (\lambda_{\max}(\mathbf{H}_m \mathbf{H}_m^T))^{\gamma/2} (\text{tr}(\mathbf{H}_l \mathbf{H}_l^T))^{\gamma/2} \leq CM^2 \bar{R}^{\gamma/2}. \end{aligned}$$

So $T_{22} = O_p(M^{2/\gamma} \bar{R}^{1/2})$. Based on Condition 2, we deduce that

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{L(\mathbf{w}) - R(\mathbf{w})}{R(\mathbf{w})} \right| \leq \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |L(\mathbf{w}) - R(\mathbf{w})| = O_p(\xi_N^{-1} N^{-1} M^{2/\gamma} \bar{R}^{1/2}) = o_p(1).$$

Thus (S.3) holds. This completes the proof.

C.3. Proof of Theorem 3

To prove Theorem 3, we need to study the difference between $R(\hat{\mathbf{w}})$ and $R(\mathbf{w}^*)$. By the proof of (S.2), under Condition 3, we derive that

$$\sup_{\mathbf{w} \in \mathcal{W}} |C_N^*(\mathbf{w}) - L(\mathbf{w})| = O_p(N^{-1} M^{1/\gamma} \bar{R}^{1/2}) = o_p(1). \quad (\text{S.4})$$

Also, according to the proof of (S.3), under Condition 3, we have

$$\sup_{\mathbf{w} \in \mathcal{W}} |L(\mathbf{w}) - R(\mathbf{w})| = O_p(N^{-1} M^{2/\gamma} \bar{R}^{1/2}) = o_p(1). \quad (\text{S.5})$$

Then, by the definition of $\mathbf{w}^*$, combining (S.4) and (S.5), it is readily seen that

$$\begin{aligned} 0 &\leq R(\hat{\mathbf{w}}) - R(\mathbf{w}^*) = R(\hat{\mathbf{w}}) - C_N^*(\mathbf{w}^*) + C_N^*(\mathbf{w}^*) - L(\mathbf{w}^*) + L(\mathbf{w}^*) - R(\mathbf{w}^*) \\ &\leq R(\hat{\mathbf{w}}) - C_N^*(\hat{\mathbf{w}}) + C_N^*(\mathbf{w}^*) - L(\mathbf{w}^*) + L(\mathbf{w}^*) - R(\mathbf{w}^*) \\ &\leq \sup_{\mathbf{w} \in \mathcal{W}} |R(\mathbf{w}) - C_N^*(\mathbf{w})| + \sup_{\mathbf{w} \in \mathcal{W}} |C_N^*(\mathbf{w}) - L(\mathbf{w})| + \sup_{\mathbf{w} \in \mathcal{W}} |L(\mathbf{w}) - R(\mathbf{w})| \\ &\leq 2 \sup_{\mathbf{w} \in \mathcal{W}} |C_N^*(\mathbf{w}) - L(\mathbf{w})| + 2 \sup_{\mathbf{w} \in \mathcal{W}} |L(\mathbf{w}) - R(\mathbf{w})| = o_p(1). \end{aligned}$$

Note that $\mathbf{w}^*$ is a well-separated minimum of $R(\mathbf{w})$, that is, for any fixed $\delta > 0$, there exists a constant $\zeta_\delta > 0$ such that $R(\mathbf{w}) - R(\mathbf{w}^*) > \zeta_\delta$ for all $\mathbf{w}$ satisfying $\|\mathbf{w} - \mathbf{w}^*\|_\infty > \delta$. This yields the event inclusion $\{\|\hat{\mathbf{w}} - \mathbf{w}^*\|_\infty > \delta\} \subseteq \{R(\hat{\mathbf{w}}) - R(\mathbf{w}^*) > \zeta_\delta\}$. Therefore,

$$P(\|\hat{\mathbf{w}} - \mathbf{w}^*\|_\infty > \delta) \leq P(R(\hat{\mathbf{w}}) - R(\mathbf{w}^*) > \zeta_\delta) \rightarrow 0$$

as $N \rightarrow \infty$. This completes the proof of Theorem 3.

C.4. Proof of Theorem 4

Let $\tilde{C}_N^*(\mathbf{w}) = N^{-1} \tilde{C}(\mathbf{w}) - N^{-1} \|\epsilon\|^2$. Because the factor N^{-1} and the term $\|\epsilon\|^2$ are unrelated to $\mathbf{w}$, we have $\tilde{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathcal{W}} \tilde{C}(\mathbf{w}) = \arg \min_{\mathbf{w} \in \mathcal{W}} \tilde{C}_N^*(\mathbf{w})$. To derive Theorem 4, it suffices to prove that

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{\tilde{C}_N^*(\mathbf{w}) - L(\mathbf{w})}{R(\mathbf{w})} \right| \xrightarrow{P} 0 \quad (\text{S.6})$$

and

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{L(\mathbf{w}) - R(\mathbf{w})}{R(\mathbf{w})} \right| \xrightarrow{P} 0. \quad (\text{S.7})$$

We first consider the proof of (S.6). Through some basic decomposition and calculation, we can derive that

$$\begin{aligned}\tilde{C}_N^*(\mathbf{w}) - L(\mathbf{w}) &= N^{-1} \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (y_i - \hat{f}_{\mathbf{w}}^j(\mathbf{x}_i))^2 - N^{-1} \sum_{i=1}^N (f_0(\mathbf{x}_i) - \hat{f}_{\mathbf{w}}(\mathbf{x}_i))^2 - N^{-1} \|\epsilon\|^2 \\ &= N^{-1} \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (\hat{f}_{\mathbf{w}}(\mathbf{x}_i) - \hat{f}_{\mathbf{w}}^j(\mathbf{x}_i))^2 + 2N^{-1} \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} \epsilon_i (f_0(\mathbf{x}_i) - \hat{f}_{\mathbf{w}}^j(\mathbf{x}_i)) \\ &\quad + 2N^{-1} \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (\hat{f}_{\mathbf{w}}(\mathbf{x}_i) - f_0(\mathbf{x}_i)) (\hat{f}_{\mathbf{w}}^j(\mathbf{x}_i) - \hat{f}_{\mathbf{w}}(\mathbf{x}_i)) \triangleq G_1(\mathbf{w}) + G_2(\mathbf{w}) + G_3(\mathbf{w}).\end{aligned}$$

Then, it is seen that

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{\tilde{C}_N^*(\mathbf{w}) - L(\mathbf{w})}{R(\mathbf{w})} \right| \leq \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_1(\mathbf{w})| + \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_2(\mathbf{w})| + \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_3(\mathbf{w})|.$$

We next show that the three terms on the right-hand side of the above display are all $o_p(1)$ one by one. It can be seen that by the Jensen's inequality, under Conditions 4 and 6,

$$\begin{aligned}\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_1(\mathbf{w})| &= \xi_N^{-1} N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} \left| \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} \left(\sum_{m=1}^M w_m \hat{f}_m(\mathbf{x}_i) - \sum_{m=1}^M w_m \hat{f}_m^j(\mathbf{x}_i) \right)^2 \right| \\ &\leq \xi_N^{-1} N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} \left| \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} \sum_{m=1}^M w_m \left(\hat{f}_m(\mathbf{x}_i) - \hat{f}_m^j(\mathbf{x}_i) \right)^2 \right| \\ &\leq \xi_N^{-1} N^{-1} \left| \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} \max_{1 \leq m \leq M} \left(\hat{f}_m(\mathbf{x}_i) - f_m^*(\mathbf{x}_i) + f_m^*(\mathbf{x}_i) - \hat{f}_m^j(\mathbf{x}_i) \right)^2 \right| = \xi_N^{-1} O_p(a_N^2) = o_p(1).\end{aligned}$$

For the term $G_2(\mathbf{w})$, we note that

$$G_2(\mathbf{w}) = 2N^{-1} \sum_{i=1}^N \sum_{m=1}^M w_m \epsilon_i (f_0(\mathbf{x}_i) - f_m^*(\mathbf{x}_i)) + 2N^{-1} \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} \sum_{m=1}^M w_m \epsilon_i (f_m^*(\mathbf{x}_i) - \hat{f}_m^j(\mathbf{x}_i)) \triangleq G_{21}(\mathbf{w}) + G_{22}(\mathbf{w}).$$

To prove $\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_2(\mathbf{w})| = o_p(1)$, we need to verify that $\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_{21}(\mathbf{w})| = o_p(1)$ and $\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_{22}(\mathbf{w})| = o_p(1)$. Using the Markov's inequality, Theorem 2 in Whittle (1960) and Conditions 1, 5–6, for any fixed $\varepsilon > 0$, we obtain that as $N \rightarrow \infty$,

$$\begin{aligned}P\left(\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_{21}(\mathbf{w})| > \varepsilon\right) &\leq P\left(\max_{1 \leq m \leq M} \left| \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} \epsilon_i (f_0(\mathbf{x}_i) - f_m^*(\mathbf{x}_i)) \right| > \varepsilon N \xi_N / 2\right) \\ &\leq 4\varepsilon^{-2} \xi_N^{-2} N^{-2} \sum_{m=1}^M E\left(\sum_{j=1}^J \sum_{i \in \mathcal{D}_j} \epsilon_i (f_0(\mathbf{x}_i) - f_m^*(\mathbf{x}_i))\right)^2 \leq C\varepsilon^{-2} \xi_N^{-2} N^{-2} \sum_{m=1}^M \sum_{i=1}^N (f_0(\mathbf{x}_i) - f_m^*(\mathbf{x}_i))^2 = O(MN^{-1}/\xi_N^2) \rightarrow 0.\end{aligned}$$

Hence, we verify that $\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_{21}(\mathbf{w})| = o_p(1)$. Further, by applying the Schwarz inequality, under Conditions 1 and 4, we can see that

$$\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_{22}(\mathbf{w})| \leq 2N^{-1} \xi_N^{-1} \max_{1 \leq m \leq M} \left(\sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (f_m^*(\mathbf{x}_i) - \hat{f}_m^j(\mathbf{x}_i))^2 \right)^{1/2} \left(\sum_{j=1}^J \sum_{i \in \mathcal{D}_j} \epsilon_i^2 \right)^{1/2}$$

$$= N^{-1} \xi_N^{-1} O_p(N^{1/2} a_N) \times O_p(N^{1/2}) = O_p(a_N N^{-1/2} / \xi_N) = o_p(1).$$

Now we finish the verification of $\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_2(\mathbf{w})| = o_p(1)$. Observe that the proof of $\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_1(\mathbf{w})| = o_p(1)$ implies $\sup_{\mathbf{w} \in \mathcal{W}} \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (\widehat{f}_{\mathbf{w}}^{-j}(\mathbf{x}_i) - \widehat{f}_{\mathbf{w}}(\mathbf{x}_i))^2 = O_p(N a_N^2)$ under Conditions 4 and 6. Denote $f_{\mathbf{w}}^*(\mathbf{x}_i) = \sum_{m=1}^M w_m f_m^*(\mathbf{x}_i)$. Then, by the Schwarz inequality, we can see that

$$\begin{aligned} \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_3(\mathbf{w})| &= 2N^{-1} \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} \left| \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (\widehat{f}_{\mathbf{w}}(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i) + f_{\mathbf{w}}^*(\mathbf{x}_i) - f_0(\mathbf{x}_i)) (\widehat{f}_{\mathbf{w}}^{-j}(\mathbf{x}_i) - \widehat{f}_{\mathbf{w}}(\mathbf{x}_i)) \right| \\ &\leq 2N^{-1} \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} \left(\sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (\widehat{f}_{\mathbf{w}}^{-j}(\mathbf{x}_i) - \widehat{f}_{\mathbf{w}}(\mathbf{x}_i))^2 \right)^{1/2} \sup_{\mathbf{w} \in \mathcal{W}} \left(\sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (\widehat{f}_{\mathbf{w}}(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i))^2 \right)^{1/2} \\ &\quad + 2N^{-1} \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} \left(\sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (\widehat{f}_{\mathbf{w}}^{-j}(\mathbf{x}_i) - \widehat{f}_{\mathbf{w}}(\mathbf{x}_i))^2 \right)^{1/2} \sup_{\mathbf{w} \in \mathcal{W}} \left(\sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (f_{\mathbf{w}}^*(\mathbf{x}_i) - f_0(\mathbf{x}_i))^2 \right)^{1/2}. \end{aligned}$$

Further, by the Jensen's inequality, we have

$$\sup_{\mathbf{w} \in \mathcal{W}} \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (\widehat{f}_{\mathbf{w}}(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i))^2 \leq \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} \max_{1 \leq m \leq M} (\widehat{f}_m(\mathbf{x}_i) - f_m^*(\mathbf{x}_i))^2 = O_p(N a_N^2). \quad (\text{S.8})$$

Based on the Jensen's inequality and Condition 5, it can be seen that

$$\sup_{\mathbf{w} \in \mathcal{W}} \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} (f_{\mathbf{w}}^*(\mathbf{x}_i) - f_0(\mathbf{x}_i))^2 \leq \sum_{j=1}^J \sum_{i \in \mathcal{D}_j} \max_{1 \leq m \leq M} (f_m^*(\mathbf{x}_i) - f_0(\mathbf{x}_i))^2 = O(N).$$

Then, with Condition 6, we have that $\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |G_3(\mathbf{w})| = O_p(\xi_N^{-1} a_N^2) + O(\xi_N^{-1} a_N) = o(1)$. This finishes the proof of (S.6).

The proof of (S.7) follows the same arguments as that of (S.3) and is therefore omitted. We complete the proof of Theorem 4.

C.5. Proof of Theorem 5

This proof focuses on the difference between $R(\widetilde{\mathbf{w}})$ and $R(\mathbf{w}^*)$. Under Condition 7, it is readily seen from the proof of (S.6) that

$$\sup_{\mathbf{w} \in \mathcal{W}} |\widetilde{C}_N^*(\mathbf{w}) - L(\mathbf{w})| = o_p(1). \quad (\text{S.9})$$

Similarly, under Condition 7, it follows from the proof of (S.7) that

$$\sup_{\mathbf{w} \in \mathcal{W}} |L(\mathbf{w}) - R(\mathbf{w})| = O_p(N^{-1} M^{2/\gamma} \bar{R}^{1/2}) = o_p(1). \quad (\text{S.10})$$

Then, according to the definitions of $\mathbf{w}^*$ and $\widetilde{\mathbf{w}}$, combining the results (S.9) and (S.10), we have

$$\begin{aligned} 0 &\leq R(\widetilde{\mathbf{w}}) - R(\mathbf{w}^*) = R(\widetilde{\mathbf{w}}) - \widetilde{C}_N^*(\mathbf{w}^*) + \widetilde{C}_N^*(\mathbf{w}^*) - L(\mathbf{w}^*) + L(\mathbf{w}^*) - R(\mathbf{w}^*) \\ &\leq R(\widetilde{\mathbf{w}}) - \widetilde{C}_N^*(\widetilde{\mathbf{w}}) + \widetilde{C}_N^*(\mathbf{w}^*) - L(\mathbf{w}^*) + L(\mathbf{w}^*) - R(\mathbf{w}^*) \\ &\leq \sup_{\mathbf{w} \in \mathcal{W}} |R(\mathbf{w}) - \widetilde{C}_N^*(\mathbf{w})| + \sup_{\mathbf{w} \in \mathcal{W}} |\widetilde{C}_N^*(\mathbf{w}) - L(\mathbf{w})| + \sup_{\mathbf{w} \in \mathcal{W}} |L(\mathbf{w}) - R(\mathbf{w})| \\ &\leq 2 \sup_{\mathbf{w} \in \mathcal{W}} |\widetilde{C}_N^*(\mathbf{w}) - L(\mathbf{w})| + 2 \sup_{\mathbf{w} \in \mathcal{W}} |L(\mathbf{w}) - R(\mathbf{w})| = o_p(1). \end{aligned}$$

Analogous to the proof of Theorem 3, for any fixed $\delta > 0$, there exists a constant $\widetilde{\zeta}_\delta > 0$ such that $R(\mathbf{w}) - R(\mathbf{w}^*) > \widetilde{\zeta}_\delta$ for all $\mathbf{w}$ satisfying $\|\mathbf{w} - \mathbf{w}^*\|_\infty > \delta$. Then we have

$$P(\|\widetilde{\mathbf{w}} - \mathbf{w}^*\|_\infty > \delta) \leq P(R(\widetilde{\mathbf{w}}) - R(\mathbf{w}^*) > \widetilde{\zeta}_\delta) \rightarrow 0$$

as $N \rightarrow \infty$. This completes the proof of Theorem 5.

C.6. Proof of Theorem 6

Let $\bar{C}_N^*(\mathbf{w}) = N^{-1} \sum_{l=1}^L \sum_{i \in S_l} (y_i - \bar{f}_{\mathbf{w}}^{-l}(\mathbf{x}_i))^2 - N^{-1} \|\epsilon\|^2$. So we have $\bar{\mathbf{w}} = \arg \min_{\mathbf{w} \in \mathcal{W}} \bar{C}(\mathbf{w}) = \arg \min_{\mathbf{w} \in \mathcal{W}} \bar{C}_N^*(\mathbf{w})$. To prove Theorem 6, it suffices to show that

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{\bar{C}_N^*(\mathbf{w}) - L_D(\mathbf{w})}{R_D(\mathbf{w})} \right| \xrightarrow{P} 0, \quad (\text{S.11})$$

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{L_D(\mathbf{w}) - R_D(\mathbf{w})}{R_D(\mathbf{w})} \right| \xrightarrow{P} 0, \quad (\text{S.12})$$

$$\text{and } \sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{R_D(\mathbf{w}) - R(\mathbf{w})}{R(\mathbf{w})} \right| \xrightarrow{P} 0. \quad (\text{S.13})$$

We first establish (S.11). It is readily shown that

$$\begin{aligned} \bar{C}_N^*(\mathbf{w}) - L_D(\mathbf{w}) &= N^{-1} \sum_{l=1}^L \sum_{i \in S_l} (y_i - \bar{f}_{\mathbf{w}}^{-l}(\mathbf{x}_i))^2 - N^{-1} \sum_{l=1}^L \sum_{i \in S_l} (f_0(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i))^2 - N^{-1} \|\epsilon\|^2 \\ &= N^{-1} \sum_{l=1}^L \sum_{i \in S_l} (\bar{f}_{\mathbf{w}}^{-l}(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i))^2 + 2N^{-1} \sum_{l=1}^L \sum_{i \in S_l} \epsilon_i (f_0(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}^{-l}(\mathbf{x}_i)) \\ &\quad + 2N^{-1} \sum_{l=1}^L \sum_{i \in S_l} (\bar{f}_{\mathbf{w}}^{-l}(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i)) (\bar{f}_{\mathbf{w}}(\mathbf{x}_i) - f_0(\mathbf{x}_i)) \triangleq S_{11}(\mathbf{w}) + S_{12}(\mathbf{w}) + S_{13}(\mathbf{w}). \end{aligned}$$

Based on Condition 9 and the Cauchy-Schwarz inequality, we have

$$\begin{aligned} |S_{11}(\mathbf{w})| &\leq N^{-1} \sum_{l=1}^L \sum_{i \in S_l} \sum_{m=1}^M w_m \left(\frac{1}{L-1} \sum_{j \neq l} \hat{f}_{m,j}(\mathbf{x}_i) - \frac{1}{L} \sum_{j=1}^L \hat{f}_{m,j}(\mathbf{x}_i) \right)^2 \\ &\leq N^{-1} \sum_{l=1}^L \sum_{i \in S_l} \max_{1 \leq m \leq M} \left(\frac{1}{L-1} \sum_{j \neq l} (\hat{f}_{m,j}(\mathbf{x}_i) - f_m^*(\mathbf{x}_i)) + \frac{1}{L} \sum_{j=1}^L (f_m^*(\mathbf{x}_i) - \hat{f}_{m,j}(\mathbf{x}_i)) \right)^2 = N^{-1} \times O_p(Na_n^2) = O_p(a_n^2). \end{aligned}$$

So under Condition 10, we can derive that $\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |S_{11}(\mathbf{w})| = O_p(a_n^2 \xi_N^{-1}) = o_p(1)$. For the term $S_{12}(\mathbf{w})$, we rewrite it as follows:

$$\begin{aligned} S_{12}(\mathbf{w}) &= 2N^{-1} \sum_{l=1}^L \sum_{i \in S_l} \epsilon_i (f_0(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i)) + 2N^{-1} \sum_{l=1}^L \sum_{i \in S_l} \epsilon_i (f_{\mathbf{w}}^*(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}^{-l}(\mathbf{x}_i)) \\ &\triangleq S_{121}(\mathbf{w}) + S_{122}(\mathbf{w}). \end{aligned}$$

Based on the Markov's inequality and Theorem 2 in Whittle (1960), under Conditions 1, 5 and 10, for any fixed $\varepsilon > 0$, we can see that as $N \rightarrow \infty$,

$$\begin{aligned} P \left(\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |S_{121}(\mathbf{w})| > \varepsilon \right) &\leq P \left(\max_{1 \leq m \leq M} \left| \sum_{l=1}^L \sum_{i \in S_l} \epsilon_i (f_0(\mathbf{x}_i) - f_m^*(\mathbf{x}_i)) \right| > \varepsilon N \xi_N / 2 \right) \\ &\leq 4\varepsilon^{-2} \xi_N^{-2} N^{-2} \sum_{m=1}^M E \left(\sum_{l=1}^L \sum_{i \in S_l} \epsilon_i (f_0(\mathbf{x}_i) - f_m^*(\mathbf{x}_i)) \right)^2 \leq C\varepsilon^{-2} \xi_N^{-2} N^{-2} \sum_{m=1}^M \sum_{i=1}^N (f_0(\mathbf{x}_i) - f_m^*(\mathbf{x}_i))^2 = O(MN^{-1}/\xi_N^2) \rightarrow 0, \end{aligned}$$

which implies that $\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |S_{121}(\mathbf{w})| = o_p(1)$. According to the fact that $\sum_{i=1}^N \epsilon_i^2 = O_p(N)$, under Conditions 1, 9 and 10, by the Schwarz inequality, it is readily shown that

$$\begin{aligned} \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |S_{122}(\mathbf{w})| &\leq 2N^{-1} \xi_N^{-1} \max_{1 \leq m \leq M} \left(\sum_{l=1}^L \sum_{i \in S_l} \left(f_m^*(\mathbf{x}_i) - \frac{1}{L-1} \sum_{j \neq l} \widehat{f}_{m,j}(\mathbf{x}_i) \right)^2 \right)^{1/2} \left(\sum_{i=1}^N \epsilon_i^2 \right)^{1/2} \\ &= N^{-1} \xi_N^{-1} O_p(N^{1/2} a_n) \times O_p(N^{1/2}) = O_p(a_n N^{-1/2} / \xi_N) = o_p(1). \end{aligned}$$

Thus we can see that $\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |S_{12}(\mathbf{w})| \leq \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |S_{121}(\mathbf{w})| + \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |S_{122}(\mathbf{w})| = o_p(1)$. Before calculating $S_{13}(\mathbf{w})$, based on Conditions 5 and 9, we can deduce the following term:

$$\sum_{l=1}^L \sum_{i \in S_l} (\bar{f}_{\mathbf{w}}(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i))^2 \leq \sum_{l=1}^L \sum_{i \in S_l} \sum_{l=1}^L \frac{1}{L} \max_{1 \leq m \leq M} (\widehat{f}_{m,l}(\mathbf{x}_i) - f_m^*(\mathbf{x}_i))^2 = O_p(N a_n^2) \quad (\text{S.14})$$

and

$$\sum_{l=1}^L \sum_{i \in S_l} (f_{\mathbf{w}}^*(\mathbf{x}_i) - f_0(\mathbf{x}_i))^2 \leq \max_{1 \leq m \leq M} \sum_{l=1}^L \sum_{i \in S_l} (f_m^*(\mathbf{x}_i) - f_0(\mathbf{x}_i))^2 = O(N).$$

Additionally, the proof of $|S_{11}(\mathbf{w})| = O_p(a_n^2)$ implies that $\sum_{l=1}^L \sum_{i \in S_l} (\bar{f}_{\mathbf{w}}^{-l}(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i))^2 = O_p(N a_n^2)$. Then, using the Schwarz inequality, we have that

$$\begin{aligned} |S_{13}(\mathbf{w})| &\leq 2N^{-1} \left(\sum_{l=1}^L \sum_{i \in S_l} (\bar{f}_{\mathbf{w}}^{-l}(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i))^2 \right)^{1/2} \left(\sum_{l=1}^L \sum_{i \in S_l} (\bar{f}_{\mathbf{w}}(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i))^2 \right)^{1/2} \\ &\quad + 2N^{-1} \left(\sum_{l=1}^L \sum_{i \in S_l} (\bar{f}_{\mathbf{w}}^{-l}(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i))^2 \right)^{1/2} \left(\sum_{l=1}^L \sum_{i \in S_l} (f_{\mathbf{w}}^*(\mathbf{x}_i) - f_0(\mathbf{x}_i))^2 \right)^{1/2} = O_p(a_n^2) + O_p(a_n). \end{aligned}$$

Therefore, it can be seen that $\xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |S_{13}(\mathbf{w})| = o_p(1)$ based on Condition 10. Further, by (S.13) (the proof is given later),

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{\bar{C}_N^*(\mathbf{w}) - L_D(\mathbf{w})}{R_D(\mathbf{w})} \right| \leq \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |S_{11}(\mathbf{w})| + \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |S_{12}(\mathbf{w})| + \xi_N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} |S_{13}(\mathbf{w})| = o_p(1).$$

This finishes the proof of (S.11).

Next, we focus on the proof of (S.12). Let $\mathbf{X}_l = (\mathbf{x}_i : i \in S_l)^T \in R^{n \times p}$ and $\mathbf{Y}_l = (y_i : i \in S_l)^T \in R^n$ denote the covariate matrix and response vector on the l th machine, respectively. Then the fitted values on the k th machine, computed by the local KRR estimator trained on the l th machine under the m th candidate model, can be written as $\widehat{f}_{m,l}(\mathbf{X}_k) = \mathbf{K}_{m,kl} (\mathbf{K}_{m,ll} + n\lambda_m \mathbf{I}_n)^{-1} \mathbf{Y}_l$. Let $\mathbf{K}_{m,l} = (\mathbf{K}_{m,1l}^T, \dots, \mathbf{K}_{m,Ll}^T)^T$. Further, denote $\mathbf{H}_{m,kl} = \mathbf{K}_{m,kl} (\mathbf{K}_{m,ll} + n\lambda_m \mathbf{I}_n)^{-1}$, $\mathbf{H}_{m,l} = \mathbf{K}_{m,l} (\mathbf{K}_{m,ll} + n\lambda_m \mathbf{I}_n)^{-1}$, $f_0(\mathbf{X}_l) = (f_0(\mathbf{x}_i) : i \in S_l)^T \in R^n$ and $\boldsymbol{\epsilon}_l = \mathbf{Y}_l - f_0(\mathbf{X}_l)$.

Initially, we derive an alternative expression for $R_{D,m}$. Let $\widehat{f}_{m,l}(\mathbf{X}_k) = (\widehat{f}_{m,l}(\mathbf{x}_i) : i \in S_k)^T \in R^n$ and $\bar{f}_m(\mathbf{X}_k) = L^{-1} \sum_{l=1}^L \widehat{f}_{m,l}(\mathbf{X}_k)$. Then,

$$\begin{aligned} R_{D,m} &= E \left(\sum_{l=1}^L \|f_0(\mathbf{X}_l) - \bar{f}_m(\mathbf{X}_l)\|^2 \right) = E \left(\sum_{l=1}^L \left\| f_0(\mathbf{X}_l) - L^{-1} \sum_{k=1}^L \mathbf{H}_{m,lk} \mathbf{Y}_k \right\|^2 \right) \\ &= E \left\| f_0(\mathbf{X}) - \frac{1}{L} \sum_{l=1}^L \mathbf{H}_{m,l} f_0(\mathbf{X}_l) - \frac{1}{L} \sum_{l=1}^L \mathbf{H}_{m,l} \boldsymbol{\epsilon}_l \right\|^2 \end{aligned}$$

$$= \left\| f_0(\mathbf{X}) - \frac{1}{L} \sum_{l=1}^L \mathbf{H}_{m,l} f_0(\mathbf{X}_l) \right\|^2 + E \left(\left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{m,l} \boldsymbol{\epsilon}_l \right)^T \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{m,l} \boldsymbol{\epsilon}_l \right) \right).$$

Similarly, we derive a convenient expression for $L_D(\mathbf{w}) - R_D(\mathbf{w})$. For $l = 1, \dots, L$, denote $\mathbf{H}_{\mathbf{w},l} = \sum_{m=1}^M w_m \mathbf{H}_{m,l}$. Then we have

$$L_D(\mathbf{w}) - R_D(\mathbf{w}) = N^{-1} \left\| f_0(\mathbf{X}) - \frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} f_0(\mathbf{X}_l) - \frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right\|^2 - R_D(\mathbf{w}) \triangleq S_{21}(\mathbf{w}) + S_{22}(\mathbf{w}),$$

where

$$S_{21}(\mathbf{w}) = 2N^{-1} \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right)^T \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} f_0(\mathbf{X}_l) - f_0(\mathbf{X}) \right)$$

and

$$S_{22}(\mathbf{w}) = \frac{1}{N} \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right)^T \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right) - E \left(\frac{1}{N} \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right)^T \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right) \right).$$

We first consider $S_{21}(\mathbf{w})$. Based on the c_r inequality, Theorem 2 in Whittle (1960) and Condition 1, we see that

$$\begin{aligned} & E \left(\sup_{\mathbf{w} \in \mathcal{W}} 2N^{-1} \left\| \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right)^T \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} f_0(\mathbf{X}_l) - f_0(\mathbf{X}) \right) \right\|^2 \right) \\ & \leq E \left(\sup_{\mathbf{w} \in \mathcal{W}} 2N^{-1} \sum_{m=1}^M \sum_{k=1}^M w_m w_k \left\| \frac{1}{L} \sum_{l=1}^L \boldsymbol{\epsilon}_l^T \mathbf{H}_{m,l}^T \left(\frac{1}{L} \sum_{r=1}^L \mathbf{H}_{k,r} f_0(\mathbf{X}_r) - f_0(\mathbf{X}) \right) \right\|^2 \right) \\ & \leq CN^{-2} \sum_{m=1}^M \sum_{k=1}^M E \left\| \frac{1}{L} \sum_{l=1}^L \boldsymbol{\epsilon}_l^T \mathbf{H}_{m,l}^T \left(\frac{1}{L} \sum_{r=1}^L \mathbf{H}_{k,r} f_0(\mathbf{X}_r) - f_0(\mathbf{X}) \right) \right\|^2 \\ & \leq CN^{-2} \sum_{m=1}^M \sum_{k=1}^M L^{-1} \sum_{l=1}^L E \left\| \boldsymbol{\epsilon}_l^T \mathbf{H}_{m,l}^T \left(\frac{1}{L} \sum_{r=1}^L \mathbf{H}_{k,r} f_0(\mathbf{X}_r) - f_0(\mathbf{X}) \right) \right\|^2 \\ & \leq CN^{-2} \sum_{m=1}^M \sum_{k=1}^M L^{-1} \sum_{l=1}^L \left\| \mathbf{H}_{m,l}^T \left(\frac{1}{L} \sum_{r=1}^L \mathbf{H}_{k,r} f_0(\mathbf{X}_r) - f_0(\mathbf{X}) \right) \right\|^2. \end{aligned}$$

Observe that $\lambda_{\max}(\mathbf{H}_{m,l} \mathbf{H}_{m,l}^T) = \lambda_{\max}((\mathbf{K}_{m,ll}) + n\lambda_m \mathbf{I}_n)^{-2} \sum_{j=1}^L \mathbf{K}_{m,jl}^T \mathbf{K}_{m,jl}$. Then under Condition 8, we have that

$$\begin{aligned} & E \left(\sup_{\mathbf{w} \in \mathcal{W}} 2N^{-1} \left\| \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right)^T \left(\frac{1}{L} \sum_{r=1}^L \mathbf{H}_{\mathbf{w},r} f_0(\mathbf{X}_r) - f_0(\mathbf{X}) \right) \right\|^2 \right) \\ & \leq CN^{-2} L \sum_{m=1}^M \sum_{k=1}^M L^{-2} \sum_{l=1}^L \lambda_{\max}(\mathbf{H}_{m,l} \mathbf{H}_{m,l}^T) \left\| \frac{1}{L} \sum_{r=1}^L \mathbf{H}_{k,r} f_0(\mathbf{X}_r) - f_0(\mathbf{X}) \right\|^2 \leq CN^{-2} M^2 L \bar{R}_D. \end{aligned}$$

Thus we can deduce that $\sup_{\mathbf{w} \in \mathcal{W}} |S_{21}(\mathbf{w})| = O_p(N^{-1} M L^{1/2} \bar{R}_D^{1/2})$. In the following, denote $\mathbf{A}_m = (\mathbf{H}_{m,1}, \dots, \mathbf{H}_{m,L})$ for $m = 1, \dots, M$. To derive the order of $S_{22}(\mathbf{w})$, we consider that

$$E \left(\sup_{\mathbf{w} \in \mathcal{W}} \frac{1}{N} \left\| \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right)^T \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right) - E \left(\left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right)^T \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{\mathbf{w},l} \boldsymbol{\epsilon}_l \right) \right) \right\|^2 \right)$$

$$\begin{aligned} &\leq E \left(\sup_{\mathbf{w} \in \mathcal{W}} \frac{1}{NL^2} \sum_{m=1}^M \sum_{k=1}^M w_m w_k \left| \sum_{l=1}^L \sum_{r=1}^L \epsilon_l^T \mathbf{H}_{m,l}^T \mathbf{H}_{m,r} \epsilon_r - E \left(\sum_{l=1}^L \sum_{r=1}^L \epsilon_l^T \mathbf{H}_{m,l}^T \mathbf{H}_{m,r} \epsilon_r \right) \right| \right)^2 \\ &\leq \frac{1}{N^2 L^4} \sum_{m=1}^M \sum_{k=1}^M E |\epsilon^T \mathbf{A}_m^T \mathbf{A}_k \epsilon - E(\epsilon^T \mathbf{A}_m^T \mathbf{A}_k \epsilon)|^2 \leq CN^{-2} L^{-4} \sum_{m=1}^M \sum_{k=1}^M \text{tr}(\mathbf{A}_k^T \mathbf{A}_m \mathbf{A}_m^T \mathbf{A}_k). \end{aligned}$$

Note that $\lambda_{\max}(\mathbf{A}_m \mathbf{A}_m^T) \leq \sum_{l=1}^L \lambda_{\max}(\mathbf{H}_{m,l} \mathbf{H}_{m,l}^T) = \sum_{l=1}^L \lambda_{\max}((\mathbf{K}_{m,ll}) + n\lambda_m \mathbf{I}_n)^{-2} \sum_{j=1}^L \mathbf{K}_{m,jl}^T \mathbf{K}_{m,jl} \leq C_K L^2$ uniformly for $m \in \{1, \dots, M\}$ under Condition 8. Moreover, we observe that the second term of $R_{m,D}$ can be rewritten as

$$E \left(\left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{m,l} \epsilon_l \right)^T \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{m,l} \epsilon_l \right) \right) = \frac{1}{L^2} E \left(\sum_{l=1}^L \epsilon_l^T \mathbf{H}_{m,l}^T \mathbf{H}_{m,l} \epsilon_l \right) = \frac{\sigma^2}{L^2} \sum_{l=1}^L \text{tr}(\mathbf{H}_{m,l}^T \mathbf{H}_{m,l}).$$

Hence $\text{tr}(\mathbf{A}_k \mathbf{A}_k^T) = \sum_{l=1}^L \text{tr}(\mathbf{H}_{k,l} \mathbf{H}_{k,l}^T) \leq CL^2 \bar{R}_D$. Then, we have that

$$\begin{aligned} &E \left(\sup_{\mathbf{w} \in \mathcal{W}} \frac{1}{N} \left| \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{w,l} \epsilon_l \right)^T \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{w,l} \epsilon_l \right) - E \left(\left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{w,l} \epsilon_l \right)^T \left(\frac{1}{L} \sum_{l=1}^L \mathbf{H}_{w,l} \epsilon_l \right) \right) \right| \right)^2 \\ &\leq CN^{-2} L^{-4} \sum_{m=1}^M \sum_{k=1}^M \lambda_{\max}(\mathbf{A}_m \mathbf{A}_m^T) \text{tr}(\mathbf{A}_k \mathbf{A}_k^T) \leq CN^{-2} M^2 \bar{R}_D, \end{aligned}$$

which implies that $\sup_{\mathbf{w} \in \mathcal{W}} |S_{22}(\mathbf{w})| = O_p(N^{-1} M \bar{R}_D^{1/2})$. Therefore, based on Condition 10 and (S.13), it can be shown that

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{L_D(\mathbf{w}) - R_D(\mathbf{w})}{R_D(\mathbf{w})} \right| \leq \frac{1}{\xi_N} \sup_{\mathbf{w} \in \mathcal{W}} |S_{21}(\mathbf{w})| + \frac{1}{\xi_N} \sup_{\mathbf{w} \in \mathcal{W}} |S_{22}(\mathbf{w})| = O_p \left(\frac{ML^{1/2} \bar{R}_D^{1/2}}{\xi_N N} \right) = o_p(1).$$

This finishes the proof of (S.12).

Finally, we prove (S.13). It is readily verified that

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{R_D(\mathbf{w}) - R(\mathbf{w})}{R(\mathbf{w})} \right| \leq \sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{R_D(\mathbf{w}) - L_D(\mathbf{w})}{R(\mathbf{w})} \right| + \sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{L_D(\mathbf{w}) - L(\mathbf{w})}{R(\mathbf{w})} \right| + \sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{L(\mathbf{w}) - R(\mathbf{w})}{R(\mathbf{w})} \right|.$$

Note that $\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{R_D(\mathbf{w}) - L_D(\mathbf{w})}{R(\mathbf{w})} \right| = o_p(1)$ can be verified in the same way of proving (S.12), and the proof of $\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{L(\mathbf{w}) - R(\mathbf{w})}{R(\mathbf{w})} \right| = o_p(1)$ has been provided (see, e.g., the equation (S.7)). Thus it is only necessary to show that $\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{L_D(\mathbf{w}) - L(\mathbf{w})}{R(\mathbf{w})} \right| = o_p(1)$. By some basic calculations, we observe that

$$\begin{aligned} L_D(\mathbf{w}) - L(\mathbf{w}) &= 2N^{-1} \sum_{i=1}^N f_0(\mathbf{x}_i) (\widehat{f}_{\mathbf{w}}(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i)) + N^{-1} \sum_{i=1}^N (\widehat{f}_{\mathbf{w}}(\mathbf{x}_i) + \bar{f}_{\mathbf{w}}(\mathbf{x}_i)) (\bar{f}_{\mathbf{w}}(\mathbf{x}_i) - \widehat{f}_{\mathbf{w}}(\mathbf{x}_i)) \\ &= 2N^{-1} \sum_{i=1}^N f_0(\mathbf{x}_i) (\widehat{f}_{\mathbf{w}}(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i)) + 2N^{-1} \sum_{i=1}^N f_{\mathbf{w}}^*(\mathbf{x}_i) (\bar{f}_{\mathbf{w}}(\mathbf{x}_i) - \widehat{f}_{\mathbf{w}}(\mathbf{x}_i)) \\ &\quad + \frac{1}{N} \sum_{i=1}^N (\widehat{f}_{\mathbf{w}}(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i)) (\bar{f}_{\mathbf{w}}(\mathbf{x}_i) - \widehat{f}_{\mathbf{w}}(\mathbf{x}_i)) + \frac{1}{N} \sum_{i=1}^N (\bar{f}_{\mathbf{w}}(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i)) (\bar{f}_{\mathbf{w}}(\mathbf{x}_i) - \widehat{f}_{\mathbf{w}}(\mathbf{x}_i)) \\ &\triangleq S_{31}(\mathbf{w}) + S_{32}(\mathbf{w}) + S_{33}(\mathbf{w}) + S_{34}(\mathbf{w}). \end{aligned}$$

Here, we first calculate the order of $\sup_{\mathbf{w} \in \mathcal{W}} \sum_{i=1}^N (\hat{f}_{\mathbf{w}}(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i))^2$. According to (S.8) and (S.14), we have that

$$\begin{aligned} \sup_{\mathbf{w} \in \mathcal{W}} \sum_{i=1}^N (\hat{f}_{\mathbf{w}}(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i))^2 &= \sup_{\mathbf{w} \in \mathcal{W}} \sum_{i=1}^N (\hat{f}_{\mathbf{w}}(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i) + f_{\mathbf{w}}^*(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i))^2 \\ &\leq \sup_{\mathbf{w} \in \mathcal{W}} \sum_{i=1}^N (\hat{f}_{\mathbf{w}}(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i))^2 + \sup_{\mathbf{w} \in \mathcal{W}} \sum_{i=1}^N (f_{\mathbf{w}}^*(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i))^2 = O_p(Na_N^2) + O_p(Na_n^2) = O_p(Na_n^2). \end{aligned}$$

Further, under Condition 5, using the Schwarz inequality, we can see that

$$\sup_{\mathbf{w} \in \mathcal{W}} |S_{31}(\mathbf{w})| \leq CN^{-1} \sup_{\mathbf{w} \in \mathcal{W}} \left(\sum_{i=1}^N f_0^2(\mathbf{x}_i) \right)^{1/2} \left(\sum_{i=1}^N (\hat{f}_{\mathbf{w}}(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i))^2 \right)^{1/2} = O_p(a_n).$$

Similarly, we also have that

$$\sup_{\mathbf{w} \in \mathcal{W}} |S_{32}(\mathbf{w})| \leq CN^{-1} \sup_{\mathbf{w} \in \mathcal{W}} \left(\sum_{i=1}^N (f_{\mathbf{w}}^*(\mathbf{x}_i))^2 \right)^{1/2} \left(\sum_{i=1}^N (\hat{f}_{\mathbf{w}}(\mathbf{x}_i) - \bar{f}_{\mathbf{w}}(\mathbf{x}_i))^2 \right)^{1/2} = O_p(a_n).$$

Moreover, based on the Schwarz inequality and (S.8),

$$\sup_{\mathbf{w} \in \mathcal{W}} |S_{33}(\mathbf{w})| \leq N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} \left(\sum_{i=1}^N (\hat{f}_{\mathbf{w}}(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i))^2 \sum_{i=1}^N (\bar{f}_{\mathbf{w}}(\mathbf{x}_i) - \hat{f}_{\mathbf{w}}(\mathbf{x}_i))^2 \right)^{1/2} = O_p(a_n a_N).$$

By (S.14), we have

$$\sup_{\mathbf{w} \in \mathcal{W}} |S_{34}(\mathbf{w})| \leq N^{-1} \sup_{\mathbf{w} \in \mathcal{W}} \left(\sum_{i=1}^N (\bar{f}_{\mathbf{w}}(\mathbf{x}_i) - f_{\mathbf{w}}^*(\mathbf{x}_i))^2 \sum_{i=1}^N (\bar{f}_{\mathbf{w}}(\mathbf{x}_i) - \hat{f}_{\mathbf{w}}(\mathbf{x}_i))^2 \right)^{1/2} = O_p(a_n^2).$$

Hence, with Condition 10, we can derive that

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{L_D(\mathbf{w}) - L(\mathbf{w})}{R(\mathbf{w})} \right| = \xi_N^{-1} \left(\sup_{\mathbf{w} \in \mathcal{W}} |S_{31}(\mathbf{w}) + S_{32}(\mathbf{w}) + S_{33}(\mathbf{w}) + S_{34}(\mathbf{w})| \right) \xrightarrow{P} 0.$$

This finishes the proof of (S.13). Further, we complete the proof of Theorem 6.

C.7. Proof of Corollary 1

Under conditions of Theorem 6, by (S.11)–(S.13), we have

$$\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{\bar{C}_N^*(\mathbf{w}) - R_D(\mathbf{w})}{R_D(\mathbf{w})} \right| = o_p(1). \quad (\text{S.15})$$

Let $\xi_{D,N} = \inf_{\mathbf{w} \in \mathcal{W}} R_D(\mathbf{w})$. By the definition of the infimum, there exists a sequence $\mathbf{w}_N \in \mathcal{W}$ such that

$$R_D(\mathbf{w}_N) \leq \xi_{D,N} + \eta_N \text{ and } \eta_N / \xi_{D,N} = o(1).$$

For any $\delta > 0$, since $R_D(\bar{\mathbf{w}}) \geq \xi_{D,N}$, we have

$$P \left(\left| \frac{\inf_{\mathbf{w} \in \mathcal{W}} R_D(\mathbf{w})}{R_D(\bar{\mathbf{w}})} - 1 \right| > \delta \right) = P \left(\frac{R_D(\bar{\mathbf{w}}) - \xi_{D,N}}{R_D(\bar{\mathbf{w}})} > \delta \right) \leq P \left(\frac{R_D(\bar{\mathbf{w}}) - R_D(\mathbf{w}_N) + \eta_N}{R_D(\bar{\mathbf{w}})} > \delta \right)$$

$$\begin{aligned}
 &= P \left(\frac{R_D(\bar{\mathbf{w}}) - \bar{C}_N^*(\bar{\mathbf{w}}) + \bar{C}_N^*(\bar{\mathbf{w}}) - \bar{C}_N^*(\mathbf{w}_N) + \bar{C}_N^*(\mathbf{w}_N) - R_D(\mathbf{w}_N) + \eta_N}{R_D(\bar{\mathbf{w}})} > \delta \right) \\
 &\leq P \left(\frac{|R_D(\bar{\mathbf{w}}) - \bar{C}_N^*(\bar{\mathbf{w}})| + |\bar{C}_N^*(\mathbf{w}_N) - R_D(\mathbf{w}_N)| + \eta_N}{R_D(\bar{\mathbf{w}})} > \delta \right),
 \end{aligned}$$

where the last inequality follows from the fact that $\bar{C}_N^*(\bar{\mathbf{w}}) \leq \bar{C}_N^*(\mathbf{w}_N)$. Furthermore,

$$\begin{aligned}
 &\frac{|R_D(\bar{\mathbf{w}}) - \bar{C}_N^*(\bar{\mathbf{w}})| + |\bar{C}_N^*(\mathbf{w}_N) - R_D(\mathbf{w}_N)| + \eta_N}{R_D(\bar{\mathbf{w}})} \\
 &\leq \sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{\bar{C}_N^*(\mathbf{w}) - R_D(\mathbf{w})}{R_D(\mathbf{w})} \right| + \sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{\bar{C}_N^*(\mathbf{w}) - R_D(\mathbf{w})}{R_D(\mathbf{w})} \right| \frac{R_D(\mathbf{w}_N)}{R_D(\bar{\mathbf{w}})} + \frac{\eta_N}{R_D(\bar{\mathbf{w}})}.
 \end{aligned}$$

Since $R_D(\bar{\mathbf{w}}) \geq \xi_{D,N}$ and $R_D(\mathbf{w}_N) \leq \xi_{D,N} + \eta_N$, we have

$$\frac{R_D(\mathbf{w}_N)}{R_D(\bar{\mathbf{w}})} \leq 1 + \frac{\eta_N}{\xi_{D,N}} \text{ and } \frac{\eta_N}{R_D(\bar{\mathbf{w}})} \leq \frac{\eta_N}{\xi_{D,N}}.$$

Therefore, by (S.15),

$$P \left(\left| \frac{\xi_{D,N}}{R_D(\bar{\mathbf{w}})} - 1 \right| > \delta \right) \leq P \left[\sup_{\mathbf{w} \in \mathcal{W}} \left| \frac{\bar{C}_N^*(\mathbf{w}) - R_D(\mathbf{w})}{R_D(\mathbf{w})} \right| \left\{ 2 + \frac{\eta_N}{\xi_{D,N}} \right\} + \frac{\eta_N}{\xi_{D,N}} > \delta \right] \rightarrow 0.$$

Thus,

$$\frac{R_D(\bar{\mathbf{w}})}{\inf_{\mathbf{w} \in \mathcal{W}} R_D(\mathbf{w})} \xrightarrow{P} 1.$$

This completes the proof.

C.8. Proof of Theorem 7

Under Conditions 1, 5, 8–9 and 11, according to the proofs of (S.11)–(S.13), we can readily prove that

$$\sup_{\mathbf{w} \in \mathcal{W}} |\bar{C}_N^*(\mathbf{w}) - R(\mathbf{w})| = o_P(1).$$

Based on the definitions of $\mathbf{w}^*$ and $\bar{\mathbf{w}}$, we have

$$\begin{aligned}
 0 &\leq R(\bar{\mathbf{w}}) - R(\mathbf{w}^*) = R(\bar{\mathbf{w}}) - \bar{C}_N^*(\mathbf{w}^*) + \bar{C}_N^*(\mathbf{w}^*) - R(\mathbf{w}^*) \\
 &\leq R(\bar{\mathbf{w}}) - \bar{C}_N^*(\bar{\mathbf{w}}) + \bar{C}_N^*(\mathbf{w}^*) - R(\mathbf{w}^*) \leq 2 \sup_{\mathbf{w} \in \mathcal{W}} |\bar{C}_N^*(\mathbf{w}) - R(\mathbf{w})| = o_P(1).
 \end{aligned}$$

Similar to the proofs of Theorems 3 and 5, for any fixed $\delta > 0$, there exists a constant $\bar{\zeta}_\delta > 0$ such that $R(\mathbf{w}) - R(\mathbf{w}^*) > \bar{\zeta}_\delta$ for all $\mathbf{w}$ satisfying $\|\mathbf{w} - \mathbf{w}^*\|_\infty > \delta$. Then we have

$$P(\|\bar{\mathbf{w}} - \mathbf{w}^*\|_\infty > \delta) \leq P(R(\bar{\mathbf{w}}) - R(\mathbf{w}^*) > \bar{\zeta}_\delta) \rightarrow 0$$

as $N \rightarrow \infty$. This completes the proof of Theorem 7.

C.9. Proof of Proposition 1

We first prove the stability bound for the distributed smoothing matrices. Recall that $\bar{\mathbf{H}}_m = L^{-1}(\mathbf{H}_{m,1}, \dots, \mathbf{H}_{m,L})$ and $\mathbf{H}_{m,l} = \mathbf{K}_{m,l}(\mathbf{K}_{m,ll} + n\lambda_m \mathbf{I}_n)^{-1}$. Therefore,

$$\|\bar{\mathbf{H}}_m\|_{\text{op}}^2 = \lambda_{\max}(\bar{\mathbf{H}}_m \bar{\mathbf{H}}_m^T) = \lambda_{\max}\left(\frac{1}{L^2} \sum_{l=1}^L \mathbf{H}_{m,l} \mathbf{H}_{m,l}^T\right) \leq \frac{1}{L^2} \sum_{l=1}^L \lambda_{\max}(\mathbf{H}_{m,l} \mathbf{H}_{m,l}^T).$$

It is readily shown that

$$\lambda_{\max}(\mathbf{H}_{m,l} \mathbf{H}_{m,l}^T) = \lambda_{\max}\left((\mathbf{K}_{m,ll} + n\lambda_m \mathbf{I}_n)^{-2} \mathbf{K}_{m,l}^T \mathbf{K}_{m,l}\right) = \lambda_{\max}\left((\mathbf{K}_{m,ll} + n\lambda_m \mathbf{I}_n)^{-2} \sum_{j=1}^L \mathbf{K}_{m,jl}^T \mathbf{K}_{m,jl}\right).$$

Thus, under Condition 8, we have

$$\|\bar{\mathbf{H}}_m\|_{\text{op}}^2 \leq \frac{1}{L^2} \sum_{l=1}^L \lambda_{\max}(\mathbf{H}_{m,l} \mathbf{H}_{m,l}^T) = \frac{1}{L^2} \sum_{l=1}^L \lambda_{\max}\left((\mathbf{K}_{m,ll} + n\lambda_m \mathbf{I}_n)^{-2} \sum_{j=1}^L \mathbf{K}_{m,jl}^T \mathbf{K}_{m,jl}\right) \leq C_K$$

uniformly for $m = 1, \dots, M$. The triangle inequality further gives

$$\frac{1}{2} \|\bar{\mathbf{H}}_m - \bar{\mathbf{H}}_k\|_{\text{op}} \leq \frac{1}{2} \left(\|\bar{\mathbf{H}}_m\|_{\text{op}} + \|\bar{\mathbf{H}}_k\|_{\text{op}} \right) \leq \sqrt{C_K}.$$

Thus, squaring both sides yields

$$\frac{1}{4} \|\bar{\mathbf{H}}_m - \bar{\mathbf{H}}_k\|_{\text{op}}^2 \leq C_K, \quad (\text{S.16})$$

which proves the first conclusion.

Next, we prove the risk bound. Let $\varrho = \max\{1, \frac{1}{2} \max_{m \neq k} \|\bar{\mathbf{H}}_m - \bar{\mathbf{H}}_k\|_{\text{op}}\}$. Clearly, $\varrho^2 \leq \max\{1, C_K\}$.

Under the fixed-design setting, each distributed candidate is an affine estimator of $f_0(\mathbf{X})$, since $\bar{\mathbf{z}}_m = \bar{\mathbf{H}}_m \mathbf{Y}$, where $\bar{\mathbf{H}}_m$ is deterministic. Therefore, the distributed candidates fall within the affine-estimation framework of Bellec (2018), with $\mathbf{A}_m = \bar{\mathbf{H}}_m$ and $\mathbf{b}_m = 0$, and the criterion $Q(\mathbf{w})$ coincides with the corresponding Q-aggregation criterion after normalization by N . Hence, if $\epsilon \sim N(0, \sigma^2 \mathbf{I}_N)$, applying Theorem 2.1 of Bellec (2018) and normalizing the Euclidean loss by N , we obtain

$$\|\bar{\mathbf{z}}(\hat{\mathbf{w}}_Q) - f_0(\mathbf{X})\|_N^2 \leq \min_{1 \leq m \leq M} \|\bar{\mathbf{z}}_m - f_0(\mathbf{X})\|_N^2 + 30\varrho^2 \sigma^2 \frac{\delta + 2 \log M}{N}$$

with probability at least $1 - \exp(-\delta)$. The corresponding expectation inequality in Theorem 2.1 of Bellec (2018) yields

$$R_D(\hat{\mathbf{w}}_Q) \leq E\left(\min_{1 \leq m \leq M} \|\bar{\mathbf{z}}_m - f_0(\mathbf{X})\|_N^2\right) + 60\varrho^2 \sigma^2 \frac{\log M}{N}.$$

Finally, we can obtain

$$R_D(\hat{\mathbf{w}}_Q) \leq \min_{1 \leq m \leq M} E(\|\bar{\mathbf{z}}_m - f_0(\mathbf{X})\|_N^2) + 60 \max\{1, C_K\} \sigma^2 \frac{\log M}{N}.$$

This completes the proof.

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